

Sang-Hyeon Park

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 Current affiliation: Agency for Defense Development, Daejeon, South Korea

EDUCATION

Hanyang University, Seoul, South Korea

Mar. 2020 – Feb. 2024

B.S. in Energy Engineering

GPA: 4.36/4.5 (3.92/4.0, 162 credits) | Major GPA: 4.38/4.5 (97 credits in major), *Summa Cum Laude* (Ranked 1st in Class)

RESEARCH INTEREST

Computational materials science / Cathodes / Solid electrolytes / Structure-Property relationship /
 Electrochemical interfaces / Diffusion and kinetics / Defect chemistry / AI for materials / Solid-state synthesis

PUBLICATIONS

4 Published / 5 Under Review / 2 In Preparation

† equal contribution, * corresponding author

[1] H Kang, H Roh, J Lee, **SH Park**, J Park, H Jeong, HK Yoon, TY Ahn* and Y Choi*, “Comparative study of Ta and Ga doping effects on $\text{Li}_7\text{La}_3\text{Zr}_{12}\text{O}_{12}$ garnet electrolytes for advanced thermal battery applications”, *RSC advances*, 15 (33), 27345-27355, (2025).

[2] **SH Park**, JH Lim, TY Ahn, HK Yoon, J Lee* and H Roh*, “First-Principles Insights into the Role of 3d Transition-Metal Substituents in Intercalating Li_xFeF_3 Cathode Materials for Lithium-Ion Batteries”, *ACS Applied Energy Materials*, 8 (19), 14712-14724, (2025).

[3] J Kim†, H Kang†, TY Ahn, **SH Park**, M Kim, SH Kang, H Roh, S Lee, and HS Kim*, “Impact reliability of multilayer stacked-cell thermal batteries using 3D FEM simulation and gas gun test validation”, *Journal of Materials Research and Technology*, 42, 5060-5072, (2026).

[4] **SH Park**, D Kang, HK Yoon, Y Choi, and J Lee*, “Data-Driven Design Strategies for Chemically Stabilizing Cathode–Coating Interfaces Employing Interpretable Machine Learning”, *Journal of Materials Chemistry A*, 14 (39), 26011-26025, (2026).

[Featured on the Back Cover]

[5] **SH Park***, C Youn, JH Lim, G Kim, HK Yoon, Y Choi, and J Lee*, “Stabilizing Interfaces in Magnesium Solid-State Batteries via High-throughput Coating Design”, *Under revision (round 2) in “Small Methods”*

[6] JH Lim, **SH Park**, C Youn, TY Ahn, H Roh, HK Yoon, and J Lee*, “Approaching Intrinsic High-Temperature Performance in Garnet-Type Solid Electrolyte (LLZTO): Microstructure and Surface-Chemistry Engineering for Advanced Salt-free Thermal Batteries”, *Under review in “Journal of Power Sources”*

[7] HR Yu, IW Lee, TY Ahn, WJ Ahn, **SH Park**, CN Im, HK Yoon, Y Choi, and YS Lee*, “Reaction-Controlled Fluorination of FeS_2 Cathodes via FeF_3 Surface Layer Formation for High-Voltage Thermal Batteries”, *Under review in “Batteries & Supercaps”*

[8] **SH Park**, H Won, D Kang, S Han, HK Yoon, and J Lee*, “Transferable descriptor-guided interfacial thermodynamics and electrochemical stability for stable solid-state alkali battery interfaces”, *Under review in “Chemical Engineering Journal”*

[9] J Kim†, **SH Park**†, H Roh, S Lee, HS Kim, and TY Ahn*, “Effects of Compressive and Structural Recovery Characteristics of Thermal-Battery Unit-Cell Pellets on Impact Reliability”, *Under review in “Journal of Materials Research and Technology”*

[10] **SH Park**†*, H Won†, J Lee, TY Ahn, and H Lee, “Theoretical investigation of competing NaFeF_3 polymorphs demonstrated by simple mechanochemical synthesis”, *Manuscript In preparation (Expected submission: Nov. 2026)*

[11] **SH Park**†, H Won†, JH Cho, CN Im, HR Yu, HK Yoon, and JS Yeo*, “Finite Element Investigation of Thermal Stability in Low Melting Electrolytes for Thermal Batteries”, *Manuscript In preparation (Expected submission: Jan. 2027)*

CONFERENCE PRESENTATIONS

[1] “Data-driven Evaluation of interfacial stability in Batteries: From First-Principles to Military Applications”

- Oral presentation in the *Korean Institute of Military Science and Technology* Spring meeting, Jeju, Republic of Korea (2026)

[2] “Thermal analysis of thermal batteries with low-melting electrolytes and materials guideline for lightweight system”

- Poster presentation in the *Korean Hydrogen and New Energy Society* Fall meeting, Jeju, Republic of Korea (2026)

[3] “First-principles investigated transition metal substitution into Li_xFeF_3 ”

- Poster presentation in the *Korean Ceramic Society* Spring meeting, Yeosu, Republic of Korea (2025)

RESEARCH EXPERIENCE

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- Agency for Defense Development (ADD)**, Daejeon, South Korea May 2024 – Present
Research Officer for National Defense, First Lieutenant, Republic of Korea Army
- Research in computational materials physics and chemistry
 - Computational investigation of structure–property relationship and defect chemistry in energy storage materials
 - First-principles thermodynamics for interfacial reactions and high-throughput study aided by machine learning
 - Research in military energy storage systems
 - High-voltage cathode (FeF_3) and low-melting eutectic (LiCl-KCl) or solid electrolytes (oxide garnets) for reserve thermal batteries
 - Powder atomic layer deposition (ALD) for thermal battery cathode materials for suppressed parasitic reactions
- Korea Institute of Science and Technology (KIST)**, Seoul, South Korea Sep. 2023 – Feb. 2024
Research Intern, Advisor: Dr. Seungho Yu
- Trained in computational approaches for analysis and design of sulfide solid electrolytes
 - Proposed the effect of isovalent dopant on phase stability of $\text{Li}_6\text{PS}_5\text{Cl}$ argyrodite through DFT calculations (VASP)

RESEARCH SKILLS

Experimental – Electrochemical spectroscopy impedance (EIS), mechanochemical synthesis, coin-cell fabrication, atomic layer deposition, and electrochemical testing/cycling

Computational – Quantum Espresso, VASP, LAMMPS, COMSOL Multiphysics, and machine learning

HONORS, AWARDS, and SCHOLARSHIPS

Honors & Awards

- Research Officer for National Defense (ROND) May 2024 – Present
 - First Lieutenant in the Republic of Korea Army
 - Selected as one of 20 research officers nationwide for STEM-based national defense research
- *Summa Cum Laude* Feb. 2024
 - Ranked 1st in Class
 - Distinguished Graduate Award for outstanding extracurricular contributions
 - Thesis: “Optimization of Poly(acrylamide) Hydrogel Electrolytes’ Mechanical Properties for the Flexible Thermogalvanic Cell”
- Dean’s List, Academic Excellence Award (Top 1% Academic Achievement) Jun. 2023
 - Hanyang University, College of Engineering
- 2nd Prize, 2023 MiliTECH Challenge Dec. 2023
 - Hosted by KAIST; Awarded by the Agency for Defense Development (Team leader, Main Presenter)
 - Project: “Semiconductor Quantum Dot-Fiber Connection Study for Wired Quantum Cryptography Communication”
- 1st Prize, National Undergraduate Chemical Engineering Design Competition Nov. 2023
 - Hosted by the Korean Institute of Chemical Engineers; Awarded by SK geocentric Inc. (Main Presenter)
 - Project: “Development of a highly activate adsorbents physically adsorbing toxic chemical agents”
- 2nd Prize, Energy and Environment Exploration Competition Nov. 2022
 - Hosted and awarded by the SAMSUNG engineering Inc. (Team leader, Main Presenter)
 - Project: “A study on the performance deterioration of membranes for pressure retarded osmosis power generation”

Scholarships

- National Science and Technology Scholarship Mar. 2022 – Feb. 2024
 - Korea Student Aid Foundation (KOSAF)
- Readership Scholarship Dec. 2023
 - Hanyang University Library, The Champion of 2023 Hanyang University Reading Quiz Competition
- Leadership Scholarship Mar. 2020 – Feb. 2022
 - Hanyang University, Student President in Department of Energy Engineering
- Academic Excellence Scholarship Aug. 2020
 - Hanyang University

PATENTS

[1] “Method for Manufacturing a Thermal Battery Using Surface Passivation Coating of Cathode Powder for a Thermal Battery and Thermal Battery Manufactured Thereby”, KR Patent, 10-3012498 (2026)

EXTRACURRICULAR ACTIVITIES

Leadership & Service

- Student President, 11th Student Council, Department of Energy Engineering, Hanyang University (2021)
- Member, Young Engineers Honor Society (YEHS), National Academy of Engineering of Korea (2022~)
- Undergraduate Mentor, Samsung Engineering Global Environmental Youth Program (2023)
- Tutor for freshmen: *General Chemistry and Laboratory I*, Hanyang University (2023)
- Mathematics Mentor for local high school students, B-Cycle Educational Mentoring Program, Seoul (2023)
- Undergraduate Reporter, Korea Energy Agency (2022)
- Selected Participant, Overseas Study Workshop, Korea Foundation for Advanced Studies (KFAS) (2026)

Sports & Arts

- President and Member, *Saebyeok* Theater Club, Hanyang University (2021-2023)
- Member, *ALSSA* Soccer Club, Department of Energy Engineering, Hanyang University (2021-2023)
- Keyboardist/Pianist, ADD Rock Band, Agency for Defense Development (2024~)

RELEVANT COURSEWORK

Materials Science, Industrial Physical Chemistry, Energy Thermodynamics, Materials Physics and Properties, Electrochemistry, Inorganic Material Chemistry, Transport Phenomena, Energy Conversion & Storage Materials, Instrumental Analysis, Artificial Intelligence and Machine Learning, Numerical Analysis, Polymer Chemistry

LANGUAGES

Korean (Native), English (TOEFL IBT 5.0/6.0, Apr. 2026), Mandarin (HSK Level 5)